Continuous variables + probability as area

1 RECOGNISE IT

- Measurement can take any value on an interval.
- Asked for an interval, tail, density or area.
- A single exact value has probability zero.

2 DECISION / METHOD

1. Name the random variable and its support.
2. Sketch/mark the requested region.
3. Use the distribution CDF or geometric area.
4. Check the answer is between 0 and 1.

3 FORMULA

```
P(a < X < b) = F(b) - F(a)
P(X > a) = 1 - F(a)
P(X = a) = 0
total area = 1
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: uniform and exponential densities, means and variances.
Still know: probability is area; density height is not automatically probability; use complements/differences.

5 EXCEL ROUTE

No universal continuous-probability function. Use the correct family CDF, then:
- left = CDF
- right = 1-CDF
- between = CDF(upper)-CDF(lower)

6 TINY WORKED EXAMPLE

A density is 0.20 on 0 to 5. Then `P(1<X<3)=width x height=2(0.20)=0.40`.

7 TOP EXAM TRAPS

- Adding point probabilities as if X were discrete.
- Treating f(x) as P(X=x).
- Forgetting the support: area outside it is zero.

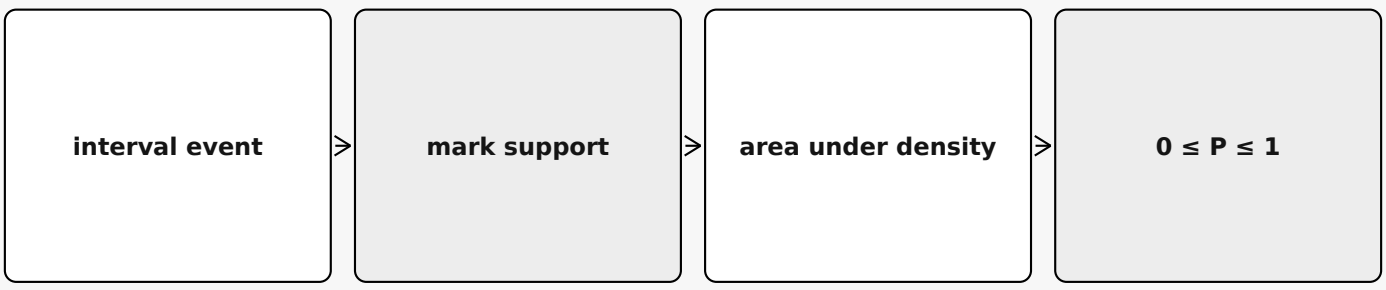
8 SAY ALOUD

For a continuous variable, probability is area over an interval, and one exact point has probability zero.

9 EXIT CHECK

Can I turn a right-tail or between event into CDF areas without looking?

ONE-GLANCE VISUAL ROUTE



Normal distribution + Z standardisation

A1S2 PRE-ASSESSMENT STUDY

1 RECOGNISE IT

- Bell-shaped, symmetric data with mean μ and SD σ.
- Course model notation is $X \sim N(\mu, \sigma^2)$.
- Need to move between original X units and standard Z units.

2 DECISION / METHOD

1. Verify X is modelled as $X \sim N(\mu, \sigma^2)$.
2. Standardise x to z using σ.
3. Use the sign as a reasonableness check.
4. Back-transform only if the answer must be in X units.

3 FORMULA

$X \sim N(\mu, \sigma^2)$

$z = (x - \mu)/\sigma$

$x = \mu + z\sigma$

$Z \sim N(0,1)$

4 FORMULA SHEET VS WHAT I KNOW

Gives: the inference table uses Z when population σ is known.
Still know: raw-score standardisation, back-transformation, and that positive z is above the mean.

5 EXCEL ROUTE

- Direct: `=NORM.DIST(x,mu,sigma,TRUE)`
- Standard: `=NORM.S.DIST(z,TRUE)`
- Do not enter variance where Excel asks for SD.

6 TINY WORKED EXAMPLE

If μ=72, σ=8 and x=84, then z=(84-72)/8=1.5: x is 1.5 SD above the mean.

7 TOP EXAM TRAPS

- COURSE NOTATION uses σ^2 as the second Normal parameter.
- EXCEL NORM.DIST asks for σ, not σ^2.
- Using SE=σ/√n for one observation X.
- Losing the sign of z.

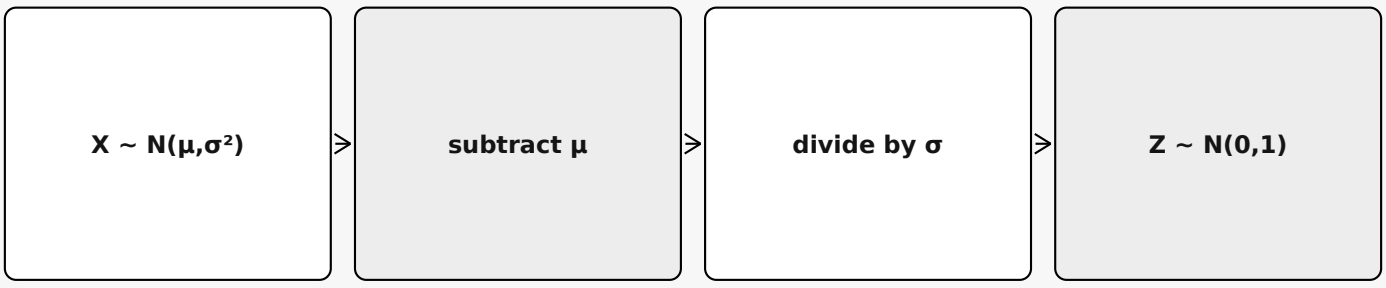
8 SAY ALOUD

Z tells me how many standard deviations the observation is from the population mean.

9 EXIT CHECK

For x below μ, what sign must z have, and why?

ONE-GLANCE VISUAL ROUTE



Normal left, right and between probabilities

1 RECOGNISE IT

- Wording says below/less than, above/more than, or between.
- X is normal and μ, σ are supplied.
- The required answer is an area/probability, not a percentile value.

2 DECISION / METHOD

1. Draw the tail(s).
2. Convert boundaries to z or use NORM.DIST directly.
3. Left: CDF. Right: 1-CDF. Between: upper CDF minus lower CDF.
4. Check area direction visually.

3 FORMULA

```
Left: P(X<x)=F(x)
Right: P(X>x)=1-F(x)
Between: P(a<X<b)=F(b)-F(a)
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: no normal tail formulas on the one-page sheet.
Still know: CDF direction, complement, and upper-minus-lower ordering.

5 EXCEL ROUTE

- `=NORM.DIST(x,mu,sigma,TRUE)`
- right: `=1-NORM.DIST(x,mu,sigma,TRUE)`
- between: two NORM.DIST calls, upper minus lower.

6 TINY WORKED EXAMPLE

For $X \sim N(50,10^2)$, `P(40<X<60)=NORM.DIST(60,50,10,TRUE)-NORM.DIST(40,50,10,TRUE)=0.6827`.

7 TOP EXAM TRAPS

- Right tail without the 1-minus.
- Between probabilities added instead of subtracted.
- Confusing X with $\bar{X}$; $\bar{X}$ would use SE.
- Course notation uses variance σ^2, while NORM.DIST uses SD σ.

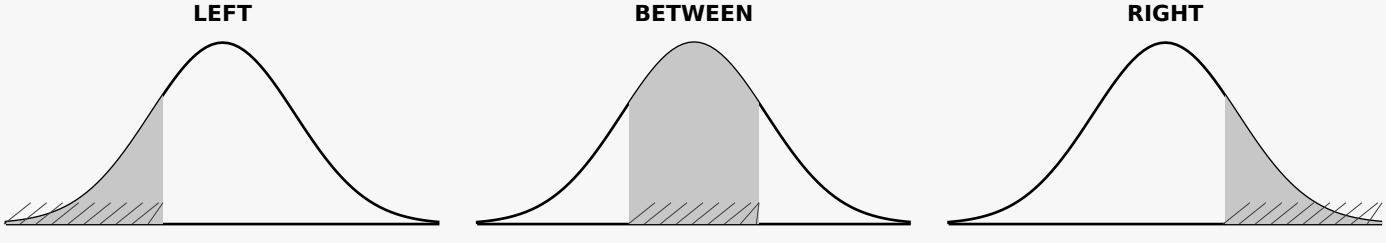
8 SAY ALOUD

Below is the CDF, above is one minus the CDF, and between is upper CDF minus lower CDF.

9 EXIT CHECK

What Excel expression gives $P(X>75)$ for $X \sim N(60,8^2)$?

ONE-GLANCE VISUAL ROUTE



Inverse normal / percentiles

1 RECOGNISE IT

- A cumulative percentage/probability is given and the unknown is x.
- Wording: percentile, cutoff, bottom q%, top q%, central C%.
- This is area -> value, not value -> area.

2 DECISION / METHOD

1. Convert wording to cumulative-left area.
2. Get z or x with an inverse function.
3. If using z, back-transform to X.
4. Check tail and sign.

3 FORMULA

```
z_q = Φ^(-1)(q)
x_q = μ + z_q*σ
top q% uses left area 1-q
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: no inverse-normal formula/function.
Still know: convert all wording to left area and back-transform from z.

5 EXCEL ROUTE

- Direct: `=NORM.INV(q,mu,sigma)`
- Via z: `=NORM.S.INV(q)`, then μ+z*σ.

6 TINY WORKED EXAMPLE

For $X \sim N(100,15^2)$, the 90th percentile is `NORM.INV(0.90,100,15)=119.22`.

7 TOP EXAM TRAPS

- Using 0.10 for the top 10% cutoff instead of 0.90.
- Using NORM.S.DIST instead of NORM.S.INV.
- Reporting z when x units are requested.
- NORM.INV asks for σ, not σ^2.

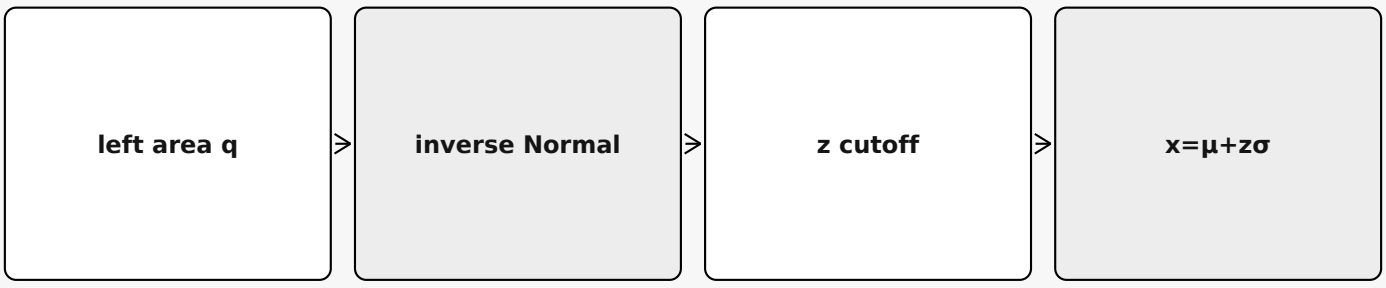
8 SAY ALOUD

Inverse normal starts with a left-tail area and returns the boundary value.

9 EXIT CHECK

What cumulative-left area represents the cutoff for the highest 5%?

ONE-GLANCE VISUAL ROUTE



Evaluating normality + normal probability plot

1 RECOGNISE IT

- Asked whether a normal model/t method is plausible.
- Raw data, histogram, boxplot or normal probability plot is supplied.
- Need an evidence-based normality judgement.

2 DECISION / METHOD

1. Check shape/symmetry and outliers.
2. Compare mean and median; skewness/kurtosis near 0 supports normality.
3. Build/read normal probability plot.
4. Approximately straight positive line -> approximately normal.

3 FORMULA

```
ordered x_(j)
alpha_j = j/(n+1)
z_j = Φ^(-1)(alpha_j)
plot x-axis z_j, y-axis ordered x_(j)
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: nothing for normality plotting.
Still know: straight-line criterion, axes, probability positions, and that one check alone is not proof.

5 EXCEL ROUTE

- Sort X ascending.
- `n=COUNT(range)`; j=1,...,n.
- α=`j/(n+1)`.
- `=NORM.S.INV(alpha)`.
- Insert XY scatter: x=Z, y=ordered X.

6 TINY WORKED EXAMPLE

With n=3 and ordered X={4,5,7}, α values are .25,.50,.75 and Z values about -0.674,0,0.674. Plot (Z,X) and inspect linearity.

7 TOP EXAM TRAPS

- Using NORM.S.DIST(α,TRUE) instead of inverse.
- Swapping axes.
- Calling any upward trend normal despite strong curvature/outliers.

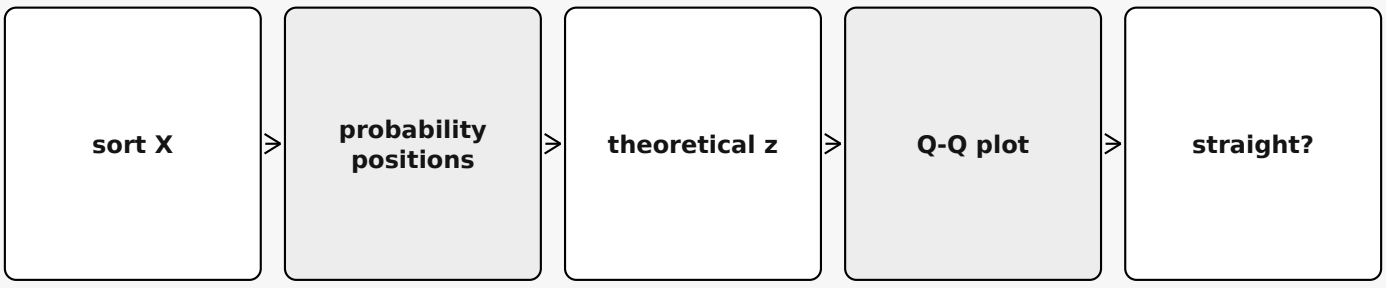
8 SAY ALOUD

Normal data make an approximately straight, positive-slope normal probability plot.

9 EXIT CHECK

Which Excel function converts alpha_j into the theoretical z_j?

ONE-GLANCE VISUAL ROUTE



Uniform distribution

1 RECOGNISE IT

- Every value between fixed bounds a and b is equally likely.
- Rectangular/constant density; zero outside [a,b].
- Probability depends only on interval length.

2 DECISION / METHOD

1. Identify support [a,b].
2. Clip the requested interval to the support.
3. Probability = overlap length / total length.
4. Use midpoint for the mean.

3 FORMULA

$f(x)=1/(b-a),\ a<=x<=b$
$P(c<X<d)=(d-c)/(b-a)$
$E(X)=(a+b)/2$
$Var(X)=(b-a)^2/12$

4 FORMULA SHEET VS WHAT I KNOW

Gives: density, expected value and variance.
Still know: interval probability as length ratio and support clipping.

5 EXCEL ROUTE

No special function needed:
`=(upper-lower)/(b-a)`
SD if needed: `=SQRT((b-a)^2/12)`.

6 TINY WORKED EXAMPLE

For X~Uniform(2,10),
$P(5<X<8)=(8-5)/(10-2)=0.375$.

7 TOP EXAM TRAPS

- Saying every interval has equal probability; only equal-length intervals do.
- Using density 1/(b-a) as the final interval probability.
- Including area outside [a,b].

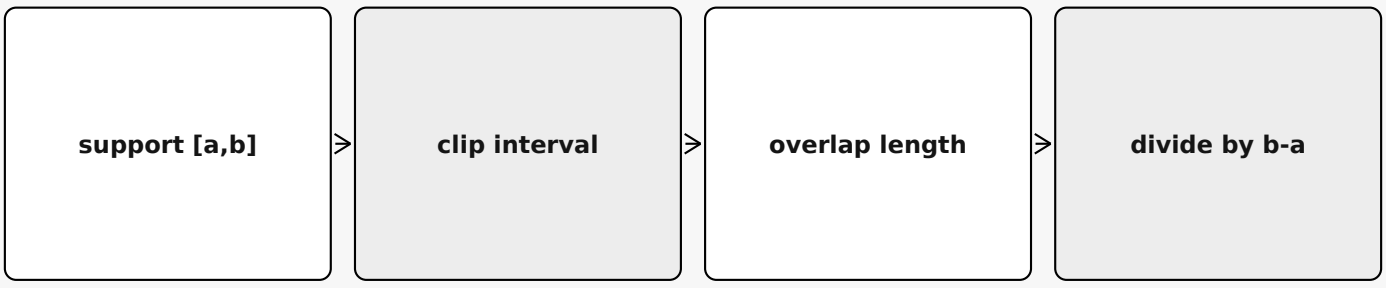
8 SAY ALOUD

Uniform probability is the requested overlap length divided by the full support length.

9 EXIT CHECK

For Uniform(-2,6), what is P(X>4)?

ONE-GLANCE VISUAL ROUTE



Exponential distribution + inverse exponential

1 RECOGNISE IT

- Waiting time/distance between random events occurring at average rate lambda.
- Right-skewed, support x>0.
- Rate and time units must match.

2 DECISION / METHOD

1. Convert rate and time to the same unit.
2. Left wait probability uses 1-exp(-lambda*t).
3. Right wait probability uses exp(-lambda*t).
4. For a percentile, solve the CDF for t.

3 FORMULA

```
f(x)=lambda*exp(-lambda*x)
F(t)=1-exp(-lambda*t)
P(X>t)=exp(-lambda*t)
t_q=-LN(1-q)/lambda
E(X)=1/lambda; Var=1/lambda^2
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: density, mean and variance.
Still know: left/right CDFs, inverse formula, memoryless context, and unit conversion.

5 EXCEL ROUTE

- Left: `=EXPON.DIST(t,lambda,TRUE)`
- Right: `=1-EXPON.DIST(t,lambda,TRUE)`
- Inverse: `=-LN(1-q)/lambda`.

6 TINY WORKED EXAMPLE

At lambda=4 per hour, `P(wait>0.25h)=EXP(-4*0.25)=0.3679`.

7 TOP EXAM TRAPS

- Mixing minutes with a per-hour rate.
- Using 1-exp(...) for a right tail.
- Treating lambda as the mean wait; mean wait is 1/lambda.

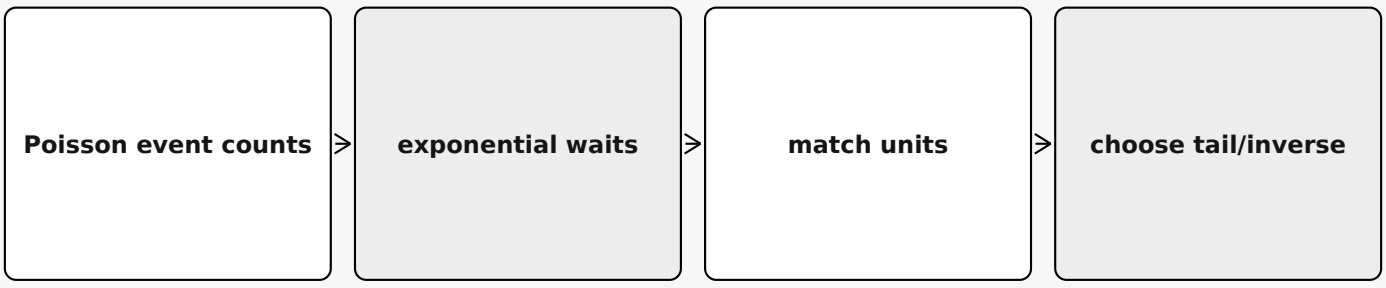
8 SAY ALOUD

Exponential waiting-time questions live or die on matching units and choosing left versus right tail.

9 EXIT CHECK

If F(t)=0.80, what equation gives the 80th-percentile wait?

ONE-GLANCE VISUAL ROUTE



Normal approximation to binomial + continuity correction

1 RECOGNISE IT

- X is a binomial count with large n.
- Exact binomial calculation is replaced by a normal model.
- Required course checks: nπ > 5 AND n(1-π) > 5.

2 DECISION / METHOD

1. Confirm nπ > 5 and n(1-π) > 5.
2. Set mean=nπ and SD=√[nπ(1-π)].
3. Apply the correct ±0.5 continuity correction.
4. Use the normal CDF/tail.

3 FORMULA

$Y \approx N(n\pi, n\pi(1-\pi))$
$SD=\sqrt{[n\pi(1-\pi)]}$
$X \le k \rightarrow Y < k+0.5$
$X \ge k \rightarrow Y > k-0.5$
$a \le X \le b \rightarrow a-0.5 < Y < b+0.5$

4 FORMULA SHEET VS WHAT I KNOW

Gives: no normal-approximation or continuity-correction rule.
Still know: conditions, normal mean/variance, and boundary shifts.

5 EXCEL ROUTE

After calculating mean and SD:
`=NORM.DIST(k+0.5,mean,SD,TRUE)`
Right tail uses 1 minus the appropriate CDF.

6 TINY WORKED EXAMPLE

If n=100, π=.40, then mean=40 and SD=sqrt(24). For P(X<=45), use `NORM.DIST(45.5,40,SQRT(24),TRUE)`.

7 TOP EXAM TRAPS

- No continuity correction.
- Wrong direction: X>k uses k+0.5, but X>=k uses k-0.5.
- Using variance where Excel wants SD.

8 SAY ALOUD

Approximate the binomial with mean n*π, SD sqrt(n*π(1-π)), then move the integer boundary by half a unit.

9 EXIT CHECK

What continuous interval represents 12<=X<=18?

ONE-GLANCE VISUAL ROUTE

